

Ihsan Ullah, Ph.D.

Tenured Associate Professor
Department of Computer Science & IT
University of Balochistan, Quetta, Pakistan.



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Employment History

- 2024 – **Chairperson.** Department of Computer Science & IT, University of Balochistan, Quetta.
- 2023 – **Tenured Associate Professor (TTS).** Department of Computer Science & IT, University of Balochistan
- 2014 – 2023 **Assistant Professor (TTS).** Department of Computer Science & IT, University of Balochistan
- 2017 – 2020 **Chairperson (In-charge).** Department of Computer Science & IT, University of Balochistan, Quetta.
- 2005 – 2014 **Lecturer.** Department of Computer Science & IT, University of Balochistan, Quetta.
- 2005 – 2005 **Lan Administrator.** Directorate of IT, University of Balochistan, Quetta.
- 2004 – 2005 **Lecturer.** on contract in Government Degree College, Sibi, Balochistan.

Education

- 2008 – 2011 **Ph.D., Université de Technologie de Troyes, France** Computer Networks
Thesis title: *Towards User-aware Peer-to-Peer Live Video Streaming Systems*
- 2007 – 2008 **Master of Science (Université de Technologie de Troyes, France** in Networks, Knowledge and Organizations.
Thesis title: *Study of the impact of user dynamics on the performance of ALM infrastructures.*
- 2002 – 2004 **Master of Computer Science**
University of Balochistan, Quetta, Pakistan
- 2001 – 2002 **Post Graduate Diploma in IT**
Government Science College Quetta, Pakistan
- 1998 – 2000 **BSc (Mathematics, Computer Science)**
Government Degree College Thana, Khyber Pakhtunkhwa, Pakistan
- 1995 – 1997 **HSSC (Pre-engineering)**
Government Degree College Mardan, Khyber Pakhtunkhwa, Pakistan

Research Publications

Journal Articles

- 1 K. Ibrahim, A. Sajid, **I. Ullah**, I. U. Khan, K. Kaushik, S. S. Askar, *et al.*, "Fuzzy inference rule based task offloading model (fi-rbtom) for edge computing," *PeerJ Computer Science*, vol. 2025, pp. 2–21, Mar. 2025, ISSN: 2167-8359. [DOI: https://doi.org/10.7717/peerj-cs.2657](https://doi.org/10.7717/peerj-cs.2657).
- 2 I. Dad, J. He, W. Noor, A. Samad, **I. Ullah**, and S. Ara, "Cross classification matrix to evaluate the performance of machine learning algorithms in predicting students performance of developing regions," *SN Computer Science*, vol. 2024, no. 5, Jun. 2024, ISSN: 2661-8907. [URL: https://doi.org/10.1007/s42979-024-02909-y](https://doi.org/10.1007/s42979-024-02909-y).

- 3 M. K. Ibrahim, A. Sajid, **I. Ullah**, T. Ali, M. Ayaz, and E.-H. M. Aggoune, "Artificial neural fuzzy inference rule-based (anfis) model for offloading tasks for edge, cloud, and uavs environment," *IEEE Access*, vol. 2024, pp. 154443–154 454, Oct. 2024, ISSN: 2169-3536.  URL: <https://ieeexplore.ieee.org/document/10721455>.
- 4 N. Khan, A. Ahmad, S. Ali, T. Jan, and **I. Ullah**, "Irs and uav-relay assisted public safety networks for video transmission: Optimizing uavs deployment and resource allocation," *Telecommunication Systems*, vol. 2024, pp. 433–449, Mar. 2024, ISSN: 1018-4864,1572-9451.  URL: <https://doi.org/10.1007/s11235-024-01128-3>.
- 5 B. Ahmad, J. Babar, M. Bakhtyar, and **I. Ullah**, "Gender based monitoring of SOP's during COVID-19 using machine learning," *Journal of Applied and Emerging Sciences*, vol. 13, no. 1, Jun. 2023, ISSN: 1814-070X,2415-2633.  URL: <https://journal.buitms.edu.pk/j/index.php/bj/article/view/585>.
- 6 F. Ahmed, W. Noor, M. A. Nasim, **I. Ullah**, and A. Basit, "Vegetation and non-vegetation classification using object detection techniques and deep learning from low/mixed resolution satellite images," *Pakistan Journal Emerging Sciences and Technologies (PJEST)*, vol. 4, no. 4, Dec. 2023, ISSN: 2788-8630.  URL: <https://doi.org/10.58619/pjest.v4i4.152>.
- 7 A. Ali, **I. Ullah**, A. Ahmed, W. Noor, A. Sajid, and I. U. Khan, "Behavior dissection of ngwn live audio and video streaming users with enhanced and efficient modelling," *Wireless Communications and Mobile Computing*, vol. 2022, pp. 1–12, Mar. 2022, ISSN: 1530-8677,1530-866.  URL: <https://www.hindawi.com/journals/wcmc/2022/7564543/>.
- 8 M. Ali, R. Asghar, **I. Ullah**, A. Ahmed, W. Noor, and J. Baber, "Towards intelligent p2p iptv overlay management through classification of peers," *Peer-to-Peer Networking and Applications*, vol. 15, no. 1, pp. 827–838, Jan. 2022, ISSN: 1936-6450, 1936-6442.  URL: <https://link.springer.com/article/10.1007/s12083-021-01288-y>.
- 9 S. Butt, M. Bakhtyar, W. Noor, J. Baber, **I. Ullah**, A. Ahmed, *et al.*, "Semantic similarity based food entities recognition using wordnet," *Journal of Intelligent & Fuzzy Systems*, vol. 43, no. 2, pp. 2069–2078, Jun. 2022, ISSN: 1875-8967,1064-1246.  URL: <https://content.iospress.com/articles/journal-of-intelligent-and-fuzzy-systems/ifs219306>.
- 10 N. u Din, A. Basit, **I. Ullah**, W. Noor, A. Ahmed, and N. Sheikh, "Brain tumor detection in mri scans using single shot multibox detector," *Journal of Intelligent & Fuzzy Systems*, vol. 43, no. 2, pp. 1985–1993, Jun. 2022, ISSN: 1875-8967,1064-1246.  URL: <https://content.iospress.com/articles/journal-of-intelligent-and-fuzzy-systems/ifs219298>.
- 11 M. J. Iqbal, **I. Ullah**, M. Ali, A. Ahmed, W. Noor, and A. Basit, "Machine learning-based stable p2p iptv overlay," *Computers, Materials & Continua*, vol. 71, no. 3, pp. 5381–5397, Jan. 2022, ISSN: 1546-2226,1546-2218.  URL: <https://www.techscience.com/cmc/v71n3/46539>.
- 12 B. Qayyum, A. Ahmed, **I. Ullah**, and S. A. Shah, "A fuzzy-logic approach for optimized and cost-effective early warning system for tsunami detection," *Sustainability*, vol. 14, no. 21, Nov. 2022, ISSN: 2071-1050.  URL: <https://www.mdpi.com/2071-1050/14/21/14516>.
- 13 A. A. Sanjrani, B. Junaid, M. Bakhtyar, **I. Ullah**, M. S. Naveed, W. Noor, *et al.*, "Extended framework for sindhi numerals ocr using gradient orientation histograms," *Journal of Intelligent & Fuzzy Systems*, vol. 43, no. 2, pp. 2045–2056, Jun. 2022, ISSN: 1875-8967,1064-1246.  URL: <https://content.iospress.com/articles/journal-of-intelligent-and-fuzzy-systems/ifs219304>.
- 14 A. Iqbal, A. Basit, I. Ali, J. Babar, and **I. Ullah**, "Automated meter reading detection using inception with single shot multi-box detector," *Intelligent Automation & Soft Computing*, vol. 27, no. 2, pp. 299–309, 2021, ISSN: 2326-005X.  DOI: 10.32604/iasc.2021.014250.
- 15 D. Munir, N. Sheikh, A. Raziq, A. Rehman, I. U. Abdul Basit, and D. Khan, "Blood pressure estimation using artificial neural network and multitaper model," *LINGUISTICA ANTVERPIENSIA*, vol. 2021,

no. 2, pp. 4041–4052, 2021, ISSN: 0304-2294.  URL:
<https://www.hivt.be/linguistica/article/view/1032>.

- 16 M. Ali, **I. Ullah**, W. Noor, A. Sajid, A. Basit, and J. Baber, “Predicting session of an iptv user through support vector regression (svr) for improved quality of streaming,” *Eng. Technol. Appl. Sci. Res*, vol. 10, no. 4, Aug. 2020.
- 17 A. Nawaz, M. Bakhtyar, J. Baber, **I. Ullah**, W. Noor, and A. Basit, “Extractive text summarization models for urdu language,” *Information Processing & Management*, vol. 57, no. 6, p. 102 383, 2020, ISSN: 1873-5371.  DOI: 10.1016/j.ipm.2020.102383.
- 18 B. Anwar, J. Baber, A. Ahmed, M. Bakhtyar, S. M. Daudpota, A. A. Sanjrani, *et al.*, “Image retrieval using visual phrases,” *International Journal of Advanced Computer Science and Applications(IJACSA)*, vol. 10, no. 3, 2019.
- 19 Z. Kanwal, A. Basit, M. Jawad, **I. Ullah**, and A. A. Sanjrani, “Overlapped apple fruit yield estimation using pixel classification and hough transform,” *International Journal of Advanced Computer Science and Applications(IJACSA)*, vol. 10, no. 2, 2019.
- 20 J. Manan, A. Ahmed, **I. Ullah**, L. Merghem-Boulahia, and D. Gaïti, “Distributed intrusion detection scheme for next generation networks,” *Journal of Network and Computer Applications*, vol. 147, p. 102 422, 2019, ISSN: 1084-8045.  DOI: 10.1016/j.jnca.2019.102422.
- 21 S. G. Shah, A. Ahmed, **I. Ullah**, and W. Noor, “A novel data aggregation scheme for wireless sensor networks,” *International Journal of Advanced Computer Science and Applications(IJACSA)*, vol. 10, no. 2, 2019.
- 22 M. Haroon, J. Baber, **I. Ullah**, S. M. Daudpota, M. Bakhtyar, and V. Devi, “Video scene detection using compact bag of visual word models,” *Advances in Multimedia*, vol. 2018, 2018.
- 23 S. Saeed, J. Baber, M. Bakhtyar, **I. Ullah**, N. Sheikh, I. Dad, *et al.*, “Empirical evaluation of svm for facial expression recognition,” *International Journal of Advanced Computer Science and Applications(IJACSA)*, vol. 9, no. 11, 2018.
- 24 A. Ahmed, R. Amin, M. R. Anjum, **I. Ullah**, and I. S. Bajwa, “Big data analytics solutions and challenges,” *Sindh University Journal*, vol. 49, no. 4, pp. 825–830, 2017.
- 25 J. Baber, M. Bakhtyar, **I. Ullah**, M. Rehman, and M. Khalid, “Effective compression of center symmetric local binary pattern,” *Mehran University Research Journal of Engineering & Technology*, vol. 36, no. 2, pp. 209–224, 2017.
- 26 V. D. Sachdeva, J. Baber, M. Bakhtyar, **I. Ullah**, W. Noor, and A. Basit, “Performance evaluation of sift and convolutional neural network for image retrieval,” *International Journal of Advanced Computer Science and Applications(IJACSA)*, vol. 8, no. 12, 2017.
- 27 J. Baber, M. Bakhtyar, W. Noor, A. Basit, and **I. Ullah**, “Performance enhancement of patch-based descriptors for image copy detection,” *International Journal of Advanced Computer Science and Applications(ijacs)*, vol. 7, no. 3, pp. 734–749, 2016.
- 28 **I. Ullah**, W. Noor, M. J. I. Khan, J. Baber, and A. Ahmed, “Challenges for a global solution addressing quality of service problems in p2p iptv systems,” *International Journal of Computer Networks and Communications Security (IJCNCS)*, vol. 3, no. 12, pp. 432–437, 2015.
- 29 **I. Ullah**, G. Doyen, G. Bonnet, and D. Gaïti, “A bayesian approach for user aware peer-to-peer video streaming systems,” *Elsevier Journal of Signal Processing: Image Communication*, vol. 27, no. 5, pp. 438–456, 2012.
- 30 **I. Ullah**, G. Doyen, G. Bonnet, and D. Gaïti, “A survey and synthesis of user behavior measurements in video streaming systems,” *IEEE Communications Surveys & Tutorials*, vol. 14, no. 3, pp. 734–749, 2012.

Books and Chapters

- 1 A. Ahmed, I. Ullah, L. Merghem-Boulahia, D. Gaiti, and G. Pujolle, "Connected innovation and technology x.o 1: Advancing digital transformation: Innovational strategies for smart industry and sustainable innovation," in *ISTE Wiley*, 2024, ch. 7, pp. 103–119.

Conference Proceedings

- 1 S. Khan, A. Ahmed, **I. Ullah**, and S. M. Zubair, "Call admission control based femtocell handover in lte networks," in *International Conference on Computing, Electronic and Electrical Engineering (ICE Cube)*, 2016, pp. 196–201.
- 2 **I. Ullah**, G. Doyen, G. Bonnet, and D. Gaïti, "Toward user-classified p2p iptv systems: A persona-based approach," in *IFIP/IEEE International Symposium on Integrated Network Management (IM)*, 2013, pp. 1187–1190.
- 3 **I. Ullah**, G. Doyen, and D. Gaïti, "Towards user-aware Peer-to-Peer live video streaming systems," in *IFIP/IEEE International Symposium on Integrated Network Management (IM)*, IEEE, 2013, pp. 920–926.
- 4 **I. Ullah**, G. Doyen, G. Bonnet, and D. Gaïti, "An autonomous topology management framework for QoS enabled P2P video streaming systems," in *International Conference on Network and Service Management (CNSM)*, IEEE, 2012, pp. 126–134.
- 5 G. Bonnet, **I. Ullah**, G. Doyen, L. Fillatre, D. Gaiti, and I. Nikiforov, "A semi-markovian individual model of users for P2P video streaming applications," in *4th IFIP International Conference on New Technologies, Mobility and Security NTMS 2011*, IEEE, 2011, pp. 1–5.
- 6 **I. Ullah**, G. Bonnet, G. Doyen, and D. Gaïti, "Un classifieur du comportement des utilisateurs dans les applications pair-à-pair de streaming vidéo," in *Colloque Francophone sur l'Ingénierie des Protocoles (CFIP 2011)*, 2011.
- 7 **I. Ullah**, G. Doyen, G. Bonnet, and D. Gaïti, "User behavior anticipation in p2p live video streaming systems through a bayesian network," in *12th IFIP/IEEE International Symposium on Integrated Network Management 2011*, IEEE, 2011, pp. 337–344.
- 8 **I. Ullah**, G. Bonnet, G. Doyen, and D. Gaïti, "Modeling user behavior in P2P live video streaming systems through a Bayesian network," in *4th IFIP International Conference on Autonomous Infrastructure, Management and Security, AIMS 2010*, Springer, 2010, pp. 2–13.
- 9 **I. Ullah**, G. Bonnet, G. Doyen, and D. Gaïti, "Improving performance of ALM systems with Bayesian estimation of peers dynamics," in *12th IFIP/IEEE International Conference on Management of Multimedia and Mobile Networks and Services, MMNS 2009*, Springer, 2009, pp. 157–169.
- 10 **I. Ullah**, G. Doyen, R. Khatoun, and D. Gaïti, "A decentralized approach to make application layer multicast systems dynamics-aware," in *10^{èmes} journées doctorales en informatique et réseaux (JDIR 2009)*, 2009, pp. 43–48.

Presentations/Trainings/Sessions

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|------|---|
| 2024 | ■ Conducted one day seminar on "Improving teaching methodologies through ICT-based solutions with lessons learned from the visit of Sapienza University of Rome, Italy" at University of Balochistan, attended both physically and online by the participants from different campuses of university of Balochistan, and other universities of the province. |
| 2024 | ■ Attended one week trainings and workshops at Sapienza University of Rome, Italy, on modes of education in the perspective of lessons learned from COVID-19 under the ERASMUS+ RAPID project. |

Presentations/Trainings/Sessions (continued)

- 2021  Conducted an awareness session on Cyber Security for the students of Univeristy of Balochistan.
- 2018  Conducted a training of the staff for the safe use of digital resources with a a focus on awareness against social engineering
- 2014  Conducted a session for the faculty members of University of Balochistan on **semester system** for university education.
- 2013  **Integrated Network Management (IM)**, 13th IFIP/IEEE International Symposium on, May 2013, **Ghent, Belgium**
- 2013  **IFIP/IEEE ManFi**, 5th International workshop on Management of the Future Internet (IFIP/IEEE ManFi, May 2013, **Ghent, Belgium**
- 2011  **Integrated Network Management (IM)**, 12th IFIP/IEEE International Symposium on Integrated Network Management (IM), May 26, 2011, **Dublin Ireland**
- 2010  **Réseaux Grand Est (RGE)**, Oct. 07, 2010, **Belfort, France**
- 2010  **4th International Conference on Autonomous Infrastructure, Management and Security (AIMS 2010)**, June 21-25, 2010, **University of Zurich, Switzerland**
- 2010  **5^e Forum des doctorants de L'UTT**, May 19, 2010, **Troyes, France**
- 2009  **Manweek 09**, 5th International Week on the Management of Networks and Services, Oct. 26-30, 2009, Telecom Italia Future Center, **Venice, Italy**
- 2009  **10ème Journées Doctorales en Informatique et Réseaux (JDIR)**, Feb. 2-4, 2009, **UTBM Belfort, France**

Additional Responsibilities

-  Working as Chairperson Department of Computer Science & IT, University of Balochistan since December 2023.
-  Worked as Chairperson Department of Computer Science & IT, University of Balochistan from October 2017 to December 2020.
-  Worked as MS/PhD program coordinator in the Department of Computer Science and IT, University of Balochistan for two years
-  Worked as Examination coordinator for more than two years in the Department of Computer Science and IT, University of Balochistan
-  Member of Departmental Semester Committee (undergraduate programs)
-  Member of Departmental Admission Committee (MS/PhD programs)
-  Member of the departmental board of studies (curriculum revision)
-  Member of Graduate Studies (MS/PhD programs) monitoring committee
-  Member of Affiliation Monitoring Committee
-  Member of Distance Learning Advisory Committee
-  Member of Semester Committee, University of Balochistan
-  Member of advisory committee for Center of Excellence for Result Based Management (CERBM)
- Member of purchase and selection committee for Government Innovation Lab. , University of Balochistan.

Courses taught

- Virtual Systems & Services (BSIT)
- Computer Networks (BSCS/BSIT))
- Data & Network Security (BSCS/BSIT))
- Information Security (BSIT)
- Internet Architecture and Protocols (BSIT)
- Computer communication and Networks (BSCS/BSIT)
- Advanced Object Oriented Programming (BSCS/BSIT))
- Advanced Networks (MSCS)
- Topics in Computer Networks (MSCS)
- Peer-to-Peer networks (PhD)
- Future Internet Architectures (PhD)

Research Supervision

PhD Thesis completed

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| Supervisor | “Enhancing the Streaming Quality of P2P IPTV Systems through Intelligent Topology Management”, by Muhammad Javid Iqbal Khan, PhD scholar at University of Balochistan. |
| Co-Supervisor | “Delay and Energy optimization for event-based sensor networks”, by Bushra Qayyum, PhD scholar at University of Balochistan. |

MS Thesis completed

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| Supervisor | “E-Health Care with Smart Hospital in Pervasive Environment”, by Laila Nawaz, University of Balochistan, Quetta |
| Supervisor | “Co-Existence of Association Frames and Data Frames in IEEE 802.11ah for IoT Applications”, by Muhammad Waseem, University of Balochistan |
| Supervisor | “Predicting activities of P2P IPTV users for improved design and management of the overlay”, by Muhammad Ali, University of Balochistan. |
| Supervisor | “Analysis and Modeling of Channel Switching in IPTV Systems”, by Muhammad Ish-tiaq, University of Balochistan. |
| Supervisor | “Analysis of User Behavior and its Impact on the Performance of P2P IPTV Systems” by Asia Ali, SBK Women’s University, Quetta. |
| Supervisor | “Stabilizing Peer-to-Peer IPTV Network Topologies for Improved Quality of Streaming”, by Sikander Shahzad, University of Balochistan, Quetta |
| Co-supervisor | “Efficiency of Machine Learning Techniques for user behavior modeling in P2P Live Video Steaming Networks”, by Rizwan Asghar, University of Balochistan, Quetta |

Skills

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| Languages | English, French, Urdu, Pashto |
| Coding | Java, Matlab, Python, $\text{\LaTeX}$, ... |
| Web Dev | HTML, css, JavaScript, Apache Web Server, Tomcat Web Server. |
| Misc. | Academic research, teaching, training, consultation, $\text{\LaTeX}$ typesetting and publishing. |

Awards and honors

- 2006  **Scholarship:** Master leading to PhD foreign scholarship by Higher Education Commission (HEC) Pakistan, in December, 2006.
- 2010  **Cisco Award:** Cisco award for the selected paper 4th International Conference on Autonomous Infrastructure, Management and Security (AIMS 2010).
- 2011  **IEEE Travel Grant:** IEEE travel grant to attend the 12th IFIP/IEEE International Symposium on Integrated Network Management (IM) 2011, held in Dublin, Ireland.
- 2013  **HEC Travel Grant:** HEC Pakistan travel grant to attend the 13th IFIP/IEEE International Symposium on Integrated Network Management (IM) 2013, held in Ghent, Belgium

Hiring experience

- 2023  **Faculty:** Participated as an expert in the selection of lecturers by the Federal Public Service Commission Pakistan.
- 2021  **IT Department:** Participated as an expert in hiring professionals for the land record system at the Revenue Department, Government of Balochistan.
- 2020  **Faculty:** Participated as an expert in hiring lecturers of computer science at Mastung campus, University of Balochistan, Quetta.
- 2020  **CERBM:** Member of the hiring committee for Center of Excellence for Result Based Management, University of Balochistan, Quetta.
- 2020  **GIL:** Member of the hiring committee for Government Innovation Lab, University of Balochistan, Quetta.

References

Available on Request